JUN-GI JANG

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DGIST E7-214, 333, Techno Jungang Daero, Hyeonpung-Eup, Dalseong-Gun, Daegu, 42988, Korea

RESEARCH INTERESTS

Data Mining, Query-Aware Data Systems, Streaming and Adaptive Computation, Applied AI

EXPERIENCE

Assistant Professor Daegu Gyeongbuk Institute of Science & Technology (DGIST) — Sep. 2025 - Present

Postdoctoral Researcher University of Illinois Urbana-Champaign (UIUC) — Sep. 2023 - Aug. 2025
Advisor: Prof. Hanghang Tong

Postdoctoral Researcher Seoul National University (SNU) — Mar. 2023 - Aug. 2023
Advisor: Prof. U Kang

Research Intern HYPERCONNECT — Jul. 2020 - Aug. 2020

EDUCATION

Seoul National University — Mar. 2017 - Feb. 2023
Ph.D. in Computer Science and Engineering
Thesis: Mining Real World Tensors via Efficient Tensor Decomposition Methods
Advisor: Prof. U Kang

Seoul National University — Mar. 2010 - Feb. 2017
B.S. in Mechanical and Aerospace Engineering;
and Computer Science and Engineering (double major)

BEST PAPER AWARDS

Best Paper Award (Honorable Mention), ICDE — May 2022
Best Paper Award (Best Research Paper), KDD — Aug. 2021
Best Paper Award (1st Place), Bigcomp — Jan. 2021

OTHER AWARDS AND HONORS

Outstanding Dissertation Award, SNU CSE — Feb. 2023
SNU BK21 Star Researcher Award, SNU BK21 — Feb. 2022
BK21 Best Graduate Student Award, SNU BK21 — Feb. 2022
Future Gauss Lecture Award, Gauss Labs — Feb. 2022
Naver Ph.D. Fellowship Award, Naver — Dec. 2021
Qualcomm Innovation Fellowship, Qualcomm — Nov. 2021
Yulchon AI Star Fellowship, Yulchon Foundation — Sep. 2021
Humantech Paper Award (Honorable Mention, lead-author), Samsung — Feb. 2018

REFEREED CONFERENCE AND JOURNAL PAPERS

28. **SCOUT: Coupling-Free Bounds for Trillion-Scale Top-k Retrieval in Sparse Tensor Factorization**
Jun-Gi Jang, Hyunsik Yoo, Ruizhong Qiu, Yinglong Xia, and Hanghang Tong.
ACM SIGMOD/PODS International Conference on Management of Data (**SIGMOD**), 2027, Huntington Beach, CA, USA. To Appear

27. **ProgNet: Program-Grounded Evidence Composition for Interpretable Graph Classification**
Minseok Jeon, Seunghyun Park, and **Jun-Gi Jang**.
ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (**KDD**), 2026, Jeju, South Korea. To Appear

26. **DuET: Dual-View Tensor-to-Topology Spectral Adapter for Enhancing Sparse Tensor Factorization**
Jun-Gi Jang, Jingrui He, Andrew J Margenot, Hanghang Tong.
ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (**KDD**), 2026, Jeju, South Korea. To Appear

25. **Fast and Accurate Domain Adaptation for Irregular and Regular Tensor Decomposition**
Junghun Kim, Ka Hyun Park, **Jun-Gi Jang**, and U Kang.
IEEE Transactions on Knowledge and Data Engineering (**TKDE**), 2026

24. **TUCKET: A Tensor Time Series Data Structure for Efficient and Accurate Factor Analysis over Time Ranges**
Ruizhong Qiu*, **Jun-Gi Jang***, Xiao Lin, Lihui Liu, Hanghang Tong (* equal contribution)
International Conference on Very Large Data Bases (**VLDB**), 2025, London, United Kingdom.

23. **Improving Group Fairness in Tensor Completion via Imbalance Mitigating Entity Augmentation**
Dawon Ahn*, **Jun-Gi Jang***, and Evangelos E. Papalexakis (* equal contribution)
The Pacific-Asia Conference on Knowledge Discovery and Data Mining (**PAKDD**), 2025, Sydney, Australia

22. **Fast and Accurate PARAFAC2 Decomposition for Time Range Queries on Irregular Tensors**
Jun-Gi Jang, Yong-chan Park, and U Kang
ACM International Conference on Information and Knowledge Management (**CIKM**), 2024, Boise, Idaho, USA.
Oral presentation, acceptance rate 347/1496 ≈ 23%.

21. **Compact Lossy Compression of Tensors via Neural Tensor-Train Decomposition**
Taehyung Kwon, Jihoon Ko, Jinhong Jung, **Jun-Gi Jang**, Kijung Shin
Knowledge and Information Systems (**KAIS**), Oct., 2024

20. **Compact Decomposition of Irregular Tensors for Data Compression: From Sparse to Dense to High-Order Tensors**
Taehyung Kwon, Jihoon Ko, Jinhong Jung, **Jun-Gi Jang**, and Kijung Shin
ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (**KDD**), 2024, Barcelona, Spain
Oral presentation, acceptance rate ≈ 20%.

19. **Fast and Accurate Domain Adaptation for Irregular Tensor Decomposition**
Junghun Kim, Ka Hyun Park, **Jun-Gi Jang**, and U Kang
ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (**KDD**), 2024, Barcelona, Spain
Oral presentation, acceptance rate ≈ 20%.

18. **Fast and Accurate Dual-Way Streaming PARAFAC2 for Irregular Tensors - Algorithm and Application**
Jun-Gi Jang, Jeongyoung Lee, Yong-chan Park, and U Kang
ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (**KDD**), 2023, Long Beach, CA, USA
Oral presentation, acceptance rate 313/1416 ≈ 22.1%.

17. **Accurate Open-set Recognition for Memory Workload**
Jun-Gi Jang, Sooyeon Shim, Vladimir Egay, Jeeyong Lee, Jongmin Park, Suhyun Chae, and U Kang

ACM Transactions on Knowledge Discovery from Data (**TKDD**), June, 2023

16. **Fast and accurate interpretation of workload classification model**
Sooyeon Shim, Doyeon Kim, **Jun-Gi Jang**, Suhyun Chae, Jeeyong Lee, and U Kang
PLOS ONE, March, 2023

15. **Accurate Bundle Matching and Generation via Multitask Learning with Partially Shared Parameters**
Hyunsik Jeon, **Jun-Gi Jang**, Taehun Kim, and U Kang
PLOS ONE, March, 2023

14. **Static and Streaming Tucker Decomposition for Dense Tensors**
Jun-Gi Jang and U Kang
ACM Transactions on Knowledge Discovery from Data (**TKDD**), Feb., 2023

13. **Falcon: Lightweight and Accurate Convolution Based on Depthwise Separable Convolution**
Jun-Gi Jang*, Chun Quan*, Hyun Dong Lee, and U Kang
Knowledge and Information Systems (**KAIS**), Jan., 2023 (* equal contribution)

12. **Accurate PARAFAC2 Decomposition for Temporal Irregular Tensors with Missing Values**
Jun-Gi Jang, Jeongyoung Lee, Jiwon Park, and U Kang
IEEE International Conference on Big Data (**BigData**), 2022, Osaka, Japan
Oral presentation, acceptance rate $122/633 \approx 19.2\%$.

11. **DPar2: Fast and Scalable PARAFAC2 Decomposition for Irregular Dense Tensors**
Jun-Gi Jang and U Kang
IEEE International Conference on Data Engineering (**ICDE**) 2022, Virtual Event
Oral presentation, acceptance rate $211/780 \approx 27.1\%$
Best Paper Award, Honorable Mention

10. **Large-scale tucker Tensor factorization for sparse and accurate decomposition**
Jun-Gi Jang*, Moonjeong Park*, Jongwuk Lee, and Lee Sael
The Journal of Supercomputing, May, 2022. (* equal contribution).

9. **Finding Key Structures in MMORPG Graph with Hierarchical Graph Summarization**
Jun-Gi Jang, Chaeheum Park, Changwon Jang, Geonsoo Kim, and U Kang
ACM Transactions on Knowledge Discovery from Data (**TKDD**), Feb., 2022

8. **Time-Aware Tensor Decomposition for Sparse Tensors**
Dawon Ahn, **Jun-Gi Jang**, and U Kang
Machine Learning, Sep., 2021

7. **Fast and Memory-Efficient Tucker Decomposition for Answering Diverse Time Range Queries**
Jun-Gi Jang and U Kang
ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (**KDD**), 2021, Virtual Event
Oral presentation, acceptance rate $238/1541 \approx 15.4\%$
Best Paper Award, Best Research Paper

6. **Fast and Accurate Partial Fourier Transform for Time Series Data**
Yong-chan Park, **Jun-Gi Jang**, and U Kang
ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (**KDD**), 2021, Virtual Event
Oral presentation, acceptance rate $238/1541 \approx 15.4\%$

5. **VEST: Very Sparse Tucker Factorization of Large-Scale Tensors**
Moonjeong Park*, **Jun-Gi Jang***, and Lee Sael

IEEE International Conference on Big Data and Smart Computing (**BigComp**), 2021, Online (* equal contribution)
🏆 **Best Paper Award, 1st Place**

4. **D-Tucker: Fast and Memory-Efficient Tucker Decomposition for Dense Tensors**
   Jun-Gi Jang and U Kang
   IEEE International Conference on Data Engineering (**ICDE**), 2020, Online
   Short, acceptance rate ≈ 32%

3. **S3CMTF: Fast, accurate, and scalable method for incomplete coupled matrix-tensor factorization**
   Dongjin Choi, **Jun-Gi Jang**, and U Kang
   PLOS ONE, June, 2019.

2. **High-Performance Tucker Factorization on Heterogeneous Platforms**
   Sejoon Oh, Namyong Park, **Jun-Gi Jang**, Lee Sael, and U Kang
   IEEE Transactions on Parallel and Distributed Systems (**TPDS**), Apr., 2019

1. **Zoom-SVD: Fast and Memory Efficient Method for Extracting Key Patterns in an Arbitrary Time Range**
   Jun-Gi Jang, Donjin Choi, Jinhong Jung, and U Kang
   ACM International Conference on Information and Knowledge Management (**CIKM**), 2018, Lingotto, Turin, Italy
   Oral presentation, acceptance rate 147/826 ≈ 17.8%

MANUSCRIPTS UNDER REVIEW

- 1 first-author manuscript currently under revision at a leading international conference.

RESEARCH GRANTS

Young Scientist Grants (Type B), NRF of South Korea — Mar. 2026 - Feb. 2031
Postdoctoral Fellowship Program, NRF of South Korea — Sep. 2023 - Aug. 2024

TEACHING EXPERIENCE

DGIST
Instructor, BE202 Introduction to Data Science @ DGIST — Spring 2026

Seoul National University
Lead T.A., M2177.004900 Theory and Lab of IoT, AI, and Big Data @ SNU — Spring 2020
T.A., M2177.004900 Theory and Lab of IoT, AI, and Big Data @ SNU — Fall 2019
T.A., M2177.004900 Theory and Lab of IoT, AI, and Big Data @ SNU — Spring 2019
T.A., M1522.001400 Introduction to Data Mining @ SNU — Spring 2018
T.A., M1522.000900 Data Structure @ SNU — Fall 2017

INVITED TALKS

The Future of Data Workshop 2023, KCC DB Society, KIISE — Jun. 2023
Korea Computer Congress 2022, KIISE — Jun. 2022
Korea Software Congress 2021, KIISE — Dec. 2021
Regular Seminar, Qatar Computing Research Institute (QCRI) — Sep. 2021
Korea Computer Congress 2020, KIISE — Jul. 2020
Korea Software Congress 2018, KIISE — Dec. 2018
Samsung AI Forum, Samsung — Sep. 2018

PROFESSIONAL SERVICES

Workshop Organizing Committees

Workshop on Interplay Between Classical Tensor Methods And Foundation Models	2026

Area Chair

KDD	2026

PC Member and Reviewer

KDD	2023 - 2026
CIKM	2025 - 2026
WWW	2026
AAAI	2024 - 2026
SDM	2024
KAIS journal	2024
TKDE journal	2024
TSP journal	2024
Machine Learning journal	2023 - 2024
TIST journal	2023
TPDS journal	2023
DAMI journal	2023

External Reviewer

TKDD	2025
NAACL	2024
DSAA	2024
KDD	2019 - 2022
WWW	2019 - 2021
ICLR	2021
NeurIPS	2020 - 2022
CIKM	2018 - 2019
ICDM	2018
WSDM	2018